

RKDF University Bhopal

Faculty Profile

Basic Information			
Name	Prof. Sanjay Jain		
Date of Birth	21/09/1975		
Designation	Professor		
Department	Electrical Engineering		
Experience	23 year		
Email ID	principal.bce2024@gmail.com		
Contact No	9826059342		
Educational Qualification			
Description	Year	University	
B.E. (Electronics & Power)	2000	Nagpur University	
M.Tech. (Energy Technology)	2011	R.G.P.V. University	
Ph.D. in Electrical Engg.	2017	Dr. C.V. Raman University Kota, Bilaspur (C.G.)	
Post-Doctoral Fellow	2023	RPI,USA & RKDF University, Bhopal	
Publications			
No. of Research Papers Published (Attach Annexure of Research papers published)	101		
No. of Books Published	NIL		
Books Chapters Published (Provide print of 1 st Page of Book)	03		
Research Scholars Supervised			
Research Program	Award	Under Supervision	Name of University
Ph.D	01	05	RKDF UNIVERSITY
PG Thesis/Dissertation/ M. tech	20	2	RKDF UNIVERSITY
Area of Expertise (100 words)			
In the integration of Basic Electrical Engineering, Electrical Machines, Power Systems, Control Systems, and Energy Conservation Management. I specialize in designing, analyzing, and optimizing electrical systems, ensuring efficient energy use, and implementing advanced control mechanisms for enhanced system performance and sustainability.			
Award and Achievement			
Name of Award		Description (With certified of Copy award)	
National			
International			
Conference/Seminar/Workshops/FDP			
Description	No.		
Conference/Seminar attended/ organized	36 (attended)		
Work shop attended/ organized	01 (attended)		
FDP Attended/ organised	01 (attended)		
Research Project			
Name of Project	Funding Agencies	Amount	

- Any other Achievement

List of Published Paper

S.No	Title of paper	Name of journal	Year	ISSN	Impact factor	Volume (Issue)
1	Impact of DG Allocation on Loss Minimization in Distribution Network: A Review	RGI International Journal of Applied Science & Technology	2011	ISSN: 2230-939x.	-	
2	Siting and Sizing of DG in Medium Primary Radial Distribution System with Enhanced Voltage Stability	Hindawi Publishing Corporation Chinese Journal of Engineering	2014	ISSN: 2314-8063	peer-reviewed	Volume-2014Article ID 518970, 9 pages
3	Evaluation Of Optimal Access Location And Capacity Of DG For Insertion In Distribution System With Condensed Power Losses Showing Improved Critical Bus Voltage Stipulation	Electrical and Electronics Engineering: An International Journal (ELELIJ)	2015	ISSN: 2200-5846	peer-reviewed	Volume-4 Issue-I February 2015
4	Improvement of Plant Efficiency Through Effective Maintenance Procedures: Detailed Analysis of Available Maintenance Procedures, Part-1	TIT International Journal of Science & Technology	2016	ISSN: 2319-6688		Volume-5 Issue-1
5	Improvement of Plant Efficiency Through Effective Maintenance Procedures: A case Study of Wind Farms Limited, Dewas, (M.P.) India, Part-II	TIT International Journal of Science & Technology	2016	ISSN: 2319-6688		Volume-5 Issue-1
6	Distributed generation deployment: State-of-the-art of distribution system planning in sustainable era	Renewable and Sustainable Energy Reviews	2017	ISSN: 1364-0321	6.79, SCImago Journal Rank- 3.12	77
7	Optimum Power Control and Quality Improvement in Hybrid Power System	International Journal of Advance Engineering and Research Development	2017	ISSN: 2348-4470	SJIF-4.72	Volume-4 Issue-1
8	Mitigation of Inrush Current of Transformer Using Voltage Sag Compensator	International Journal of Advance Research ,Ideas and Innovations in Technology	2017	ISSN: 2454-132X	4.295	Volume-3 Issue-1
9	Importance of Energy Conservation in Power System Economy	TIT International Journal of Science & Technology	2017	ISSN: 2319-6688		Volume-6 Issue-I
10	Trends of Economic	TIT International	2017	ISSN:		Volume-6

	Growth and Energy Demand in India	Journal of Science & Technology		2319-6688		Issue-I
11	Application of Energy Management System in Power System Analysis: Investigations of Different Modules-Part I	TIT International Journal of Science & Technology	2017	ISSN: 2319-6688		Volume-6 Issue-II
12	Application of Energy Management System in Power System Analysis: Investigations of Different Modules-Part II	TIT International Journal of Science & Technology	2017	ISSN: 2319-6688		Volume-6 Issue-II
13	A Comparative for Fault Location Estimation for Bipolar HVDC Transmission Line Using Different ANN Technique	International Journal of Contemporary Technology and Management	2018	ISSN: 2278-8034		Volume-7 Issue-V
14	A Review on HVDC Transmission Line Fault Location Detection With Different ANN Technique	International Journal of Software & Hardware Research in Engineering	2018	ISSN: 2347-4890		Volume-6 Issue-6
15	Fuzzy Logic Approach for Unit Commitment of Load Dispatching	SHODH SANGAM A RKDF University Journal of Science and Engineering	2018	ISSN: 2581-5806		Volume-1 Issue-2
16	A Survey on Energy Saving Approaches for Locomotive DC Drive System	SHODH SANGAM A RKDF University Journal of Science and Engineering	2018	ISSN: 2581-5806		Volume-1 Issue-1
17	Methods of Unit Commitment Constraints of Load	SHODH SANGAM A RKDF University Journal of Science and Engineering	2019	ISSN: 2581-5806		Volume-2 Issue-2
18	Performance Improvement of Induction Motor by Using Fuzzy Logic Controller	SHODH SANGAM A RKDF University Journal of Science and Engineering	2019	ISSN: 2581-5806		Volume-2 Issue-3
19	Speed Control of Three Phase Induction Motor by Using PI and PID Controller	SHODH SANGAM {A RKDF University Journal of Science and Engineering	2019	ISSN: 2581-5806		Volume-2 Issue-3
20	A Wide Area Damping Controller of Power System	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2019	ISSN: 2581-5806		Volume-2 No-3
21	Wide Area Damping Controller of Power System With Emphasis on Damping of Critical Inter Area Mode	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2019	ISSN: 2581-5806		Volume-2 No-3
22	Performance Improvement of Solid State Transformer	Research Journal of Engineering Technology and Management	2019	ISSN: 2582-0028		Volume-2 Issue-4
23	Transformer Less Grid Connected PV System with Reduced THD	SHODH SANGAM { A RKDF University Journal of Science and	2019	ISSN: 2581-5806		Volume-2 No-6

		Engineering				
24	The Review of Hybrid Energy Storage Topologies for Power System Application	Research Journal of Engineering Technology and Management	2019	ISSN: 2582-0028		-
25	Enhanced Loadability and Inapt Locations	International Journal of Recent Technology and Engineering (IJRTE)	2019	ISSN: 2277-3878		-
26	Fault location Method Using Fuzzy Logic membership Function	2nd International Conference on Contemporary Technological Solution towards fulfillment of Social Needs	2019	ISSN: 2581-5806		-
27	Partial Discharge Inception Rating of a Three Core Power Cable from Consideration of Discharges in Voids	2nd International Conference on Contemporary Technological Solution towards fulfillment of -Social Needs	2019	ISSN: 2581-5806		-
28	Hybrid Energy Storage System Schemes for Power System Application	Engpapper.com	-	-		-
29	The Review of Hybrid Energy Storage Topologies for Power System Application	Research Journal of Engineering Technology and Management	Special Issue 1 (2019)	ISSN: 2582-0028		-
30	MATLAB Based Digital Analysis of Partial Discharge in Power System	2nd International Conference on Contemporary Technological Solution towards fulfillment of Social Needs	2019	ISSN: 2581-5806		-
31	Partial Discharge Detection Methods in High Voltage Power Capacitor Equipment	2nd International Conference on Contemporary Technological Solution towards fulfillment of Social Needs	2019	ISSN: 2581-5806		-
32	Hybrid Energy Storage System Employing Battery And Super Capacitor To Cater Transient Load	2nd International Conference on Contemporary Technological Solution towards fulfillment of Social Needs	2019	ISSN: 2581-5806		
33	Smart Meter Data Monitoring	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
34	Brilliant Meter Using GSM Technology	Research Journal of Engineering Technology and Management	2020	ISSN: 2582-0028		-
35	H-bridge multilevel inverter for PV/Wind hybrid energy system	Research Journal of Engineering Technology and Management	March 2020	ISSN: 2582-0028		Volume 03, Issue 01

36	Five level CHB Using Pulse Width Modulation (PWM) Technique	Research Journal of Engineering Technology and Management	March 2020	ISSN: 2582-0028		Volume 03, Issue 01
37	Various levels of Multilevel Inverters and comparative Analysis	SHODH SANGAM { A RKDF University Journal of Science and Engineering	May 2020	ISSN: 2581-5806		Vol.-03, No.-03,
38	7-level cascaded H-bridge inverter using selective harmonic elimination	SHODH SANGAM { A RKDF University Journal of Science and Engineering	May 2020	ISSN: 2581-5806		Vol.-03, No.-03,
39	Performance Improvement of Doubly Fed Induction Generator (DFIG) phasor model by ANN using MATLAB/ Simulink	SHODH SANGAM { A RKDF University Journal of Science and Engineering	July 2020	ISSN: 2581-5806		Vol.-03, No.-04,
40	Artificial Neural Network (ANN) control strategy for Doubly Fed Induction Generator (DFIG)	Research Journal of Engineering Technology and Management	June 2020	ISSN: 2582-0028		Volume 03, Issue 02
41	A Literature Review on Energy Saving Approach for Locomotive DC Motor Using Power IGBTs	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
42	A Simulation Analysis of Highly Efficient Technique for Renewing Decelerating DC Drive System	International Research Journal of Technology and Applied Science	2020	ISSN: 2396-9889		-
43	Voltage Sag Compensator Based on a Transformer-Coupled Series Connected Voltage Source Inverter	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
44	The Mitigation Techniques Inrush Current of Transformer and Voltage Sag Compensator	Research Journal of Engineering Technology and Management	2020	ISSN: 2582-0028		-
45	An Exploratory Survey on Sliding Mode Controller for Induction Motor Drives	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
46	A Highly Efficient Design Mode Controller for Application of Induction Motor Drives	International Research Journal of Technology and Applied Science	2020	ISSN: 2396-9889		-
47	Hybrid Power Generation System Application for Remote Area	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
48	Cost Analysis of PV or Wind Hybrid Power Generation	Research Journal of Engineering Technology and Medical Science	2020	ISSN: 2582-6212		-
49	Designing a Charge Controller with a Maximum Power Point Tracker Using a Particle	SHODH SANGAM { A RKDF University Journal of Science and	2021	ISSN: 2581-5806		-

	Swarm Optimization Method	Engineering				
50	Performance Improvement of a Hybrid Wind-Solar Hybrid Simulated with a PI Controller	Research Journal of Engineering Technology and Medical Science	2020	ISSN: 2582-6212		-
51	A Review of Simulation - and Hardware Design of a Fly Back Converter for Solar Energy Powered DC Loads	SHODH SANGAM { A RKDF University Journal of Science and Engineering	2020	ISSN: 2581-5806		-
52	Simulation of Three-Phase Matrix Converter	International Journal for Research and Development	2020	ISSN: 2321-0613		-
53	Simulation and Hardware Design of a Fly back Converter for Solar Energy Powered DC Loads	International Journal of Scientific Research in Engineering and Management	2021	ISSN: 2582-3930		-
54	Performance enhancement of grid connected inverter with input of two PV array under different environment conditions using ANFIS control	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Oct 2021	ISSN: 2581-5806		
55	A Review on PV Inverter based on Mismatched Environmental Conditions	International journal for Research in Engineering and Emerging Trends	Aug 2021	ISSN: 2545-4523		
56	Generator–Grid Interfaces for Wind Energy Systems using Matrix Converter	SHODH SANGAM -- A RKDF University Journal of Science and Engineering	Aug 2021	ISSN: 2581-5806		
57	Latest Scenario of World Energy Demand, Electricity Consumption and Their Development Trend	Research Journal of Engineering Technology and Medical Science	Dec 2021	ISSN: 2582-6212		Volume-4 Issue-4
58	Optimal Design of Hybrid Renewable Energy System (HRES) For Off Grid Applications in Remote Areas	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec 2021	ISSN: 2581-5806		Vol.-06, No.-04,
59	Impacts of PV penetration on the performance of Power Network	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec 2021	ISSN: 2581-5806		Vol.-04, No.-06,
60	Performance Analysis of Solar system installation on load handling capability of Power system by using PSSE Simulator	Research Journal of Engineering Technology and Medical Science	Dec 2021	ISSN: 2582-6212		Volume-4 Issue-4
61	An Efficient Design of Passive High Pass Filter for Shunt Active Power Filter Applications	Research Journal of Engineering Technology and Medical Science	Dec 2021	ISSN: 2582-6212		Volume-4 Issue-4
62	A Survey on Hybrid Active Power Filter Topologies and Controllers	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec 2021	ISSN: 2581-5806		Vol.-04, No.-06,

63	Analysis of Matrix Converter in variable frequency mode	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Feb 2022	ISSN: 2581-5806		Vol.-05, No.-01,
64	MATLAB Simulink Modeling of Matrix Converter	Research Journal of Engineering Technology and Medical Science	March 2022	ISSN: 2582-6212		Volume-5 Issue-1
65	Grid Connected PV System Using Single Phase, Single Stage Current Source Inverter	Research Journal of Engineering Technology and Medical Science	March 2022	ISSN: 2582-6212		Volume-5 Issue-1
66	Multilevel Inverter with Predictive Control for Renewable Energy Smart Grid Applications	International Journal of Electrical and Electronics Research (IJEER)	2022	e-ISSN: 2347-470X	Scopus Index Journal	Volume 10 Issue 3
67	Enhancement of Power Quality with Increased Levels of MLI in Smart Grid Applications	Samriddhi – JPSET	2022	Online ISSN: 2454-5767 Print ISSN: 2229-7111	UGC-Care (Journal no. 64061).	volume 14(4)
68	Predictive Switching Control for Multilevel Inverter using CNN-LSTM for Voltage Regulation	ADBU journal of Engineering Technology (AJET)	July-2022	ISSN: 2348-7305	UGC Care	
69	Critical Analysis on Multilevel Inverter Designs for Renewable Energy Grid”	Samriddhi – JPSET,	2022.	Online ISSN: 2454-5767 Print ISSN: 2229-7111	UGC-Care (Journal no. 64061).	volume 14(3),
70	Implementation of Perturb & Observe Method in MPPT for Enhanced output of PV System	SHODH SANGAM { A RKDF University Journal of Science and Engineering	April 2022	ISSN: 2581-5806		Volume-5 Issue-2
71	Power Quality Enhancement of Distribution System by Using D-STATCOM	SHODH SANGAM { A RKDF University Journal of Science and Engineering	June 2022	ISSN: 2581-5806		Volume-5 Issue-3
72	Performance Analysis of Power System after Implementation of FACTS Devices	Research Journal of Engineering Technology and Medical Science	June 2022	ISSN: 2582-6212		Volume-5 Issue-2
73	A Review on Efficient Mitigation Techniques for Issues in Power Quality	SHODH SANGAM { A RKDF University Journal of Science and Engineering	June 2022	ISSN: 2581-5806		Volume-5 Issue-3
74	Hybrid power system for reducing energy storage requirements	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Oct 2022	ISSN: 2581-5806		Volume 11, Issue5, page 1
75	MPPT algorithm for solar wind system using standard perturb and observe method	Research Journal of Engineering Technology and Medical Science	Sept-2022	ISSN: 2582-6212		Volume 05, Issue 03
76	A Detailed Review on Efficient Mitigation	Research Journal of Engineering	June-22	ISSN:		Volume 05, Issue 02

	Techniques for Issues Concerning Power Quality	Technology and Medical Science		2582-6212		
77	Simulation and Design Analysis of DSTATCOM and DVR for improving power Quality in Distribution Network	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Oct-2022	ISSN: 2581-5806		Volume 05, Issue 5
78	Pulse-width modulation technique for three level bridges are reducing harmonic content and efficiency	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Oct-2022	ISSN: 2581-5806		Volume 05, Issue 5
79	Extraction of maximum power from SPV Arrays (SPVA)	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec-2022	ISSN: 2581-5806		Volume 05, No 6
80	Novel fault detection algorithms and evaluation of their deployment in multiple existing PV installations	Research Journal of Engineering Technology and Medical Science	Dec-22	ISSN: 2582-6212		Volume 05, Issue 04
81	A series connected Dynamic Voltage Restorer Used in Power Distribution Network	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec-2022	ISSN: 2581-5806		Volume 05, No 6
82	Analysis of Three Topologies of Dynamic Voltage Restorer to Mitigate Power Quality Issues	Research Journal of Engineering Technology and Medical Science	Dec-22	ISSN: 2582-6212		Volume 05, Issue 04
83	PV Source based Multi Area Interconnected Efficient Power System	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Dec-2022	ISSN: 2581-5806		Volume 05, No 6
84	Multi-level inverters for Renewables and Industrial Drives Applications	Research Journal of Engineering Technology and Medical Science	Dec-22	ISSN: 2582-6212		Volume 05, Issue 04
85	The AC-DC bridge converter simulated in MATLAB software	Research Journal of Engineering Technology and Medical Science	Dec-22	ISSN: 2582-6212		Volume 05, Issue 03
86	A Review on Power System Reliability with Advanced Power Quality Techniques	Research Journal of Engineering Technology and Medical Science	Sep-23	ISSN: 2582-6212		Volume 06, Issue 03
87	Particle Swarm Optimization based multi objective approach for installation of dispersed generator	Journal of Integrated Science and Technology	12-April-23	ISSN: 2321-4635	Scopus Index Journal	
88	Soft Computing based optimization employed for efficient Renewable Energy resource deployment	Journal of Integrated Science and Technology	28-Jul-2023	ISSN: 2321-4635	Scopus Index Journal	
89	Distributed generation optimization with HOMER software and fuzzy logic control	Journal of Integrated Science and Technology	28-Sep-2023	ISSN: 2321-4635	Scopus Index Journal	

90	A multi indexed performance assessment function to facilitate efficient installation of Renewable Energy resource	Journal of Integrated Science and Technology	24-Oct-2023	ISSN: 2321-4635	Scopus Index Journal	
91	A Survey on the Impact of solar controllers on power system	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Oct-2023	ISSN: 2581-5806		Vol-6, no-5
92	Efficiency Performance of Two MPPT Processes for PV System with Difference ; solar Panels Irradiances	Research Journal of Engineering Technology and Medical Science	June-2023	ISSN: 2582-6212		Vol-6, Issue-02
93	Efficient Implementaion of Improvement in Power Metrics Using Unified Conditioner Power Quality	SHODH SANGAM { A RKDF University Journal of Science and Engineering	June-2023	ISSN: 2581-5806		Vol-6, no-3
94	Power System Sustainability Enhancement Through Capacitor Placement	Process Integration and Optimization for Sustainability	12-Jan-2024	e-ISSN 2509-4246 P-ISSN 2509-4238	Springer	Vol.:(0123456789) 1 3
95	AC-DC bidirectional microgrid system for PV system application	Research Journal of Engineering Technology and Medical Science	March-2024	ISSN: 2582-6212		Vol-7, Issue-01
96	PV-fuel cell system hybrid combination with super capacitor support	Research Journal of Engineering Technology and Medical Science	March-2024	ISSN: 2582-6212		Vol-7, Issue-01
97	Optimizing The Accuracy Of 1P-3P Fault Identification Systems Using DT-ML Algorithms	Research Journal of Engineering Technology and Medical Science	March-2024	ISSN: 2582-6212		Vol-7, Issue-01
98	Analysis of PV-Fuel Cell System with Super Capacitor Support	SHODH SANGAM { A RKDF University Journal of Science and Engineering	Feb-2024	ISSN: 2581-5806		Vol-7, no-1
99	DT-ML Algorithm Based Optimization of 1P-3P Fault Identification System Accuracy	SHODH SANGAM { A RKDF University Journal of Science and Engineering	April-2024	ISSN: 2581-5806		Vol-7, no-2
100	Significance of Five Level B inverter for improving power quality of Microgrid	Research Journal of Engineering Technology and Medical Science	June-2024	ISSN: 2582-6212		Vol-7, Issue-01
101	5-Level inverter for a grid-connected PV system	SHODH SANGAM { A RKDF University Journal of Science and Engineering				

Book Chapter:

SN	Title	Year	Name of Book	Publisher	ISBN No
1	Chapter 2B: Existing Problems related to Electrical Distribution Network Part-2	2021	Active Electrical Distribution Network: A Smart Approach	Wiley Publishing Group	9781119599593
2	Enhanced Loadability and Inapt Locations Investigation in the Renewable Energy Resource Enriched Power System (Pages 845-877)	2022	Planning of Hybrid Renewable Energy Systems, Electric Vehicles and Microgrid (Modeling, Control and Optimization)	Springer Singapore	978-19-09795
3	Impact of E-Waste on the Environment (page 74-89)	2023	Sustainable Approaches and Strategies for E-Waste Management and Utilization	IGI Global	978-1-668-47573-7